Note

In the above example, the control variable is **i**, which is initialized to 10. Next the condition **i<=15** is tested**,** if the value is true **i** gets printed, then the control variable i gets updated as **i=i+1** (this can also be written as i +=1 using shorthand assignment operator). When **i** becomes **16**, the condition is returned as False and this will terminate the loop.

Note

print can have end, sep as parameters. *end* parameter can be used when we need to give any escape sequences like '\t' for tab, '\n' for new line and so on. *sep* as parameter can be used to specify any special characters like, (comma) ; (semicolon) as separator between values (Recall the concept which you have learnt in previous chapter about the formatting options in print()).

Following is an example for using else part within while loop.

Example 6.7: program to illustrate the use of while loop - with else part

```
i=10                                          # intializing part of the control variable
while (i<=15):                                # test condition
      print (i,end='\t')                      # statements - block 1
      i=i+1                                   # Updation of the control variable
else:
    print ("\nValue of i when the loop exit ",i)
```

Output: 1

```
10   11   12   13   14   15
Value of i when the loop exit  16
```

(ii) for loop

The for loop is usually known as a definite loop, because the programmer knows exactly how many times the loop will be executed.

Syntax:

```
for counter_variable in sequence:
      statements-block 1
[else:                          # optional block
      statements-block 2]
```

The for .... in statement is a looping statement used in Python to iterate over a sequence of objects, i.e., it goes through each item in a sequence. Here the sequence is the collection of ordered or unordered values or even a string.

12th Computer Science_EM Chapter 6.indd 76 26-12-2022 16:44:46